London Ambulance Service NHS Trust

Support the management of trauma in the Emergency and Urgent Care Setting

Component 8

Abdominal Injuries

Abdominal Injuries - Component 8

Objectives

- Understand signs and symptoms, complications and treatment for abdominal injuries.
- Explain when interventions from a clinician are required.
- Describe time critical injuries and when action should be taken.

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Trauma of the abdomen can be extremely difficult to assess.

- There is little protection of organs and structures in the abdomen
- Identifying that abdominal trauma has occurred is more important than identifying what structure is affected
- There may be few initial indications that a serious injury has occurred

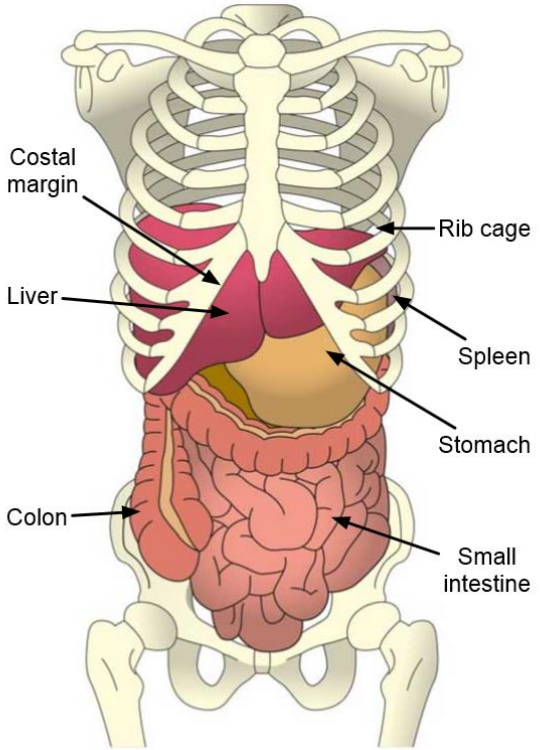


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The three anatomical areas of the abdomen are: :

- Abdominal cavity
- Pelvis
- Retro-peritoneal area



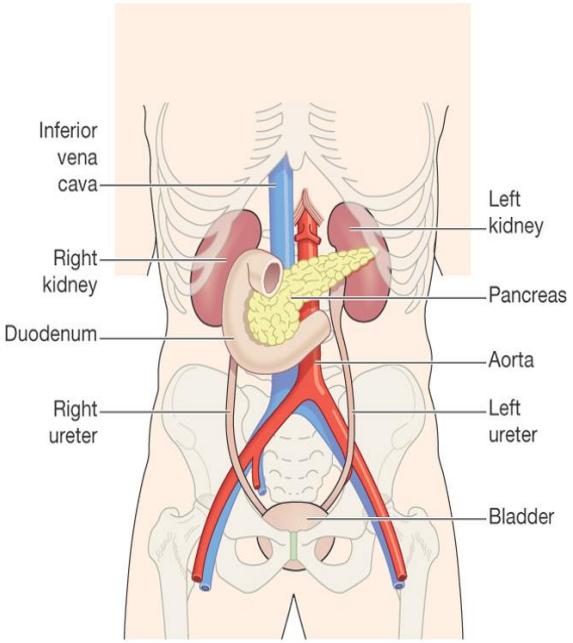
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Retro-peritoneal Area

The retro-peritoneal area lies against the posterior abdominal wall and contains:

- Kidneys and ureters, pancreas, abdominal aorta, vena cava and part of the duodenum.
- The structures are attached to the posterior abdominal wall, and are often injured by shearing, due to rapid forces.



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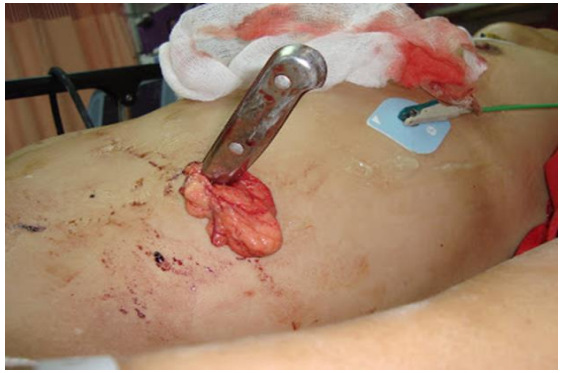
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Types of abdominal injuries

- **Blunt**



- **Penetrating**



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Blunt Injuries

- The most common pattern of abdominal injury seen
- Caused by direct blows, rapid deceleration or blast injuries and results in a closed abdominal injury



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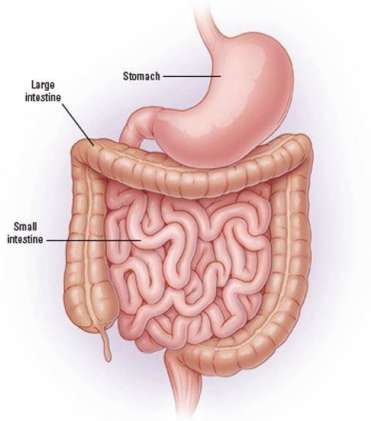
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Structures that can be injured:



- Spleen
- Liver
- Small bowel
- Duodenum
- Aorta
- Tethered structures are the most commonly injured





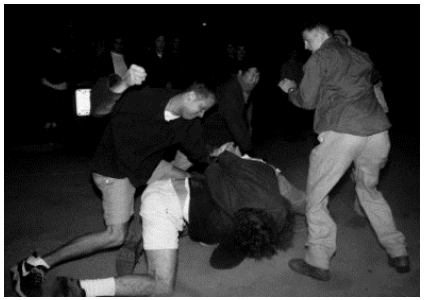
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Examples of causes of blunt trauma:



- Assault without a weapon
- Assault with a weapon
- Falls
- RTC's
- Shockwaves from a blast





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Type of injuries caused by blunt trauma:



- Internal bleeding
- Ruptured organ(s) (e.g. spleen)
- Diaphragmatic injury
- Vascular injury (e.g. aorta)
- Fractured ribs
- Be aware of pregnancy



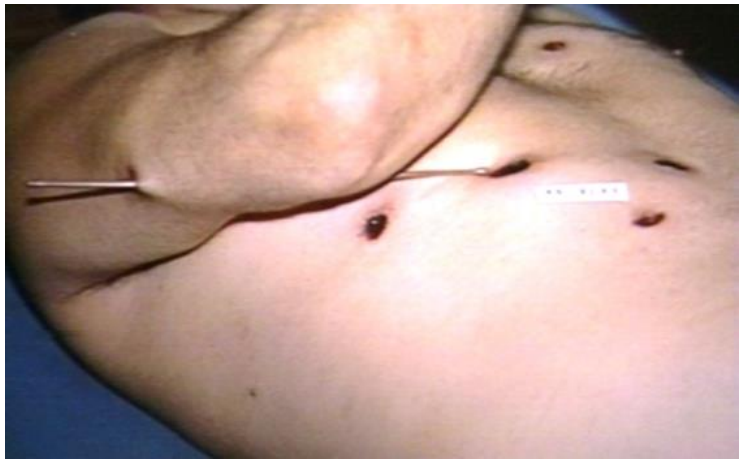
You do not have X-Ray eyes – you will not be able to diagnose the injury!

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Penetrating Injuries

- Results in an open abdominal injury
- May cause injury to adjacent structures/organs, such as the chest
- Higher risk of death than blunt trauma



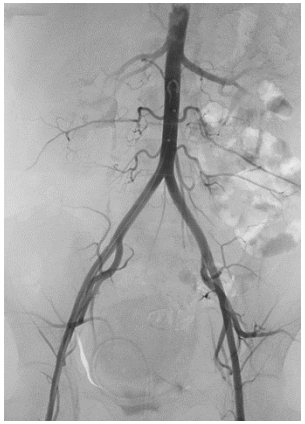
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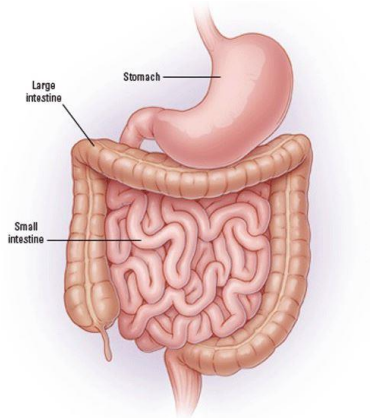
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Structures that can be injured:



- Spleen
- Liver
- Small bowel
- Duodenum
- Aorta
- Lungs
- Heart





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Causes of penetrating trauma include:

- Assault with weapons
- RTC
- Blast material/shrapnel
- Gunshot wound
- Evisceration injury







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Examples of injuries caused:

- Internal bleeding
- Ruptured organs (e.g. spleen)
- Diaphragmatic injury
- Vascular injury (e.g. aorta)
- Genitalia injuries (be aware of pregnancy)



You do not have X-Ray eyes – you will not be able to diagnose the injury!

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Do not remove penetrating object if still in situ, secure the object before transport, but allow movement if the object is pulsating.

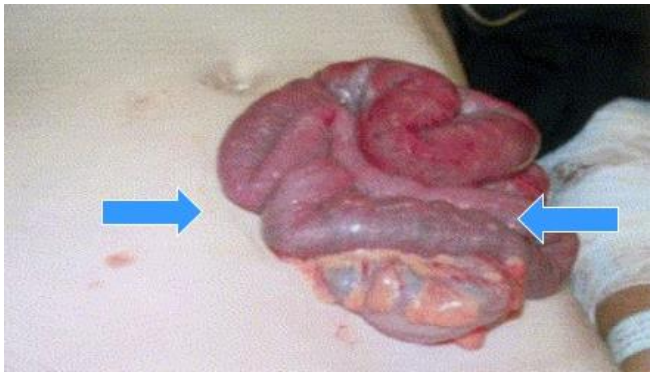


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Evisceration Injuries

Evisceration is the protrusion of abdominal organs outside the body





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Management evisceration injuries:

- Do not push any protruding organs back into the abdominal cavity

- Apply appropriate dressing
- Do not wet the dressing- warm dressing become cold dressings

Dressing: **Blast dressing**

- Place the plastic sheet against organ
- Apply the dressing to the wound
- Secure using the conforming bandage





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Important considerations:

- Personal safety – DANGER!
- Ensure police have declared the scene as safe



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Mechanism of Injury - MOI

Remember to ascertain the MOI, giving you a very important guide to the likelihood of significant injuries :

- Stabbing – Consider length of penetrating object/ Angle of insertion/Has it entered the thoracic cavity.
- Gunshot – Type and range of firearms/ Distance from firearm to victim/ Is there an entrance and an exit wound.
- RTC – Look at impact speed/use of seat belts/Vehicle damage.

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Assessment

- Assess C ABCDE
- Is there a catastrophic haemorrhage
- Is there an airway issue
- Is there a breathing issue
- Is there a circulation issue
- Does the patient require other interventions - request APP, HEMS, Paramedic, ASAP

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Assessment

- Is the patient TIME CRITICAL – Refer to Major Trauma Tool
- Is there a disability issue
- Expose and Examine Abdomen – Looking for bruising, swelling, lacerations, evisceration, impaled objects.
- Ascertain MOI

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History Taking - Pain

Correct management of abdominal conditions requires good assessment of history of pain:

- **S**ite of pain
- **O**nset and whether this is a new pain
- **C**haracter
- **R**adiates to other areas
- **A**ssociated symptoms i.e. Nausea, dizziness
- **T**iming/duration
- **E**xacerbating or relieving factors
- **S**everity

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Secondary Survey

- Done enroute to hospital if patient is time critical
- Assess for all other injuries (top to toe examination) think of associated areas – Chest, Spinal, Pelvic
- Pain Score
- Thorough review of MOI



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Management

- Administer Oxygen as per JRCALC Plus guidelines
- Analgesia as required (**AAP's can only give Entonox)**
- Monitor Vital signs – ECG If required
- Apply relevant dressings if required
- Transport to appropriate hospital





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Significant intra-abdominal trauma may not show signs immediately

Do not rule out significant injury based on vital signs alone



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Red Flags- Warning Signs



- Catastrophic haemorrhage
- Major primary survey problems
- Decreased LOC
- Shoulder tip pain
- Tachycardia
- RR <10 or >30
- Oxygen saturations <94%
- Hypotension
- Abdominal distension and bruising (very late sign)



Be aware of compensatory mechanisms

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Complications

- Infection
- Multiple organs and structures in the abdomen – unable to distinguish what has been injured
- Secondary injuries
- Patient movement with object still in situ



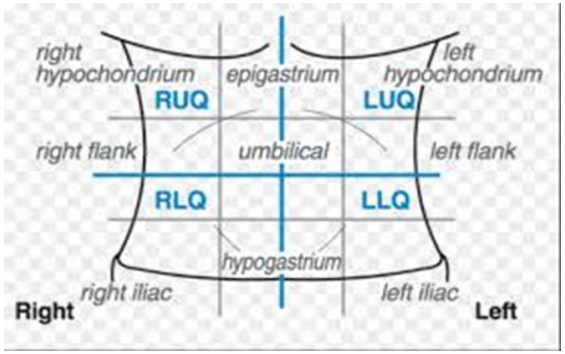


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Any Questions?





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